

Towards Morphology-Aware World Models: Extending DreamerV3 with a Kinematic Self-Model for Robotics

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This paper presents an early-stage integration of the Free-Form Kinematic Self-Model (FFKSM) into the DreamerV3 reinforcement learning framework, with the goal of equipping robotic agents with explicit morphological self-awareness during policy learning. The proposed architecture augments the actor, critic, reward predictor, and continue predictor with a 32-dimensional latent representation derived from the FFKSM, extending the DreamerV3 full state from 768 to 800 dimensions. Two experiments were conducted on the MuJoCo *Pusher-v5* environment: one with a shared third-person camera for both the world model and the self-model, and one with a decoupled ego-centric camera for the world model and a separate external camera for the self-model. Results show that the current integration does not yet yield consistent improvements over the baseline, with a representational mismatch between continuous SM latents and discrete WM latents identified as the central limiting factor. Despite this, indirect evidence suggests that the SM latents do influence agent behavior. Several directions for future work are outlined, including a learned latent encoder and an alternative architecture in which the self-model is integrated into the world model rather than the actor.

Contents

1	Introduction	3
2	Background	3
2.1	DreamerV3	3
2.2	Free-Form Kinematic Self-Model	4
3	Methodology	5
3.1	Integration Overview	6
3.2	Extraction of Self-Model Latents	6
3.3	Augmented Prediction Heads	7
3.4	Image Preprocessing for the Self-Model	7
3.5	Training Procedure	8
3.6	Damage Detection and Recovery	8
4	Experiments	9
4.1	Environment	9
4.2	Experiment 1: Shared Camera	9
4.3	Experiment 2: Decoupled Cameras	9
5	Results and Discussion	10
5.1	Benchmark	10
5.2	Experiment 1: Shared Camera	10
5.3	Experiment 2: Decoupled Cameras	11
6	Future Work	12

1 Introduction

A fundamental challenge in applying deep reinforcement learning to real-world robotics is the gap between the rich, controllable conditions of simulation and the unpredictable, partially observable nature of physical deployments. In simulation, agents typically benefit from clean global observations, known kinematics, and stable environments. In the real world, a robot may have access only to an ego-centric camera, may sustain physical damage that alters its morphology, and must act reliably under these constraints without human intervention.

World model-based reinforcement learning, as exemplified by DreamerV3 [2], addresses part of this challenge by learning a compact latent model of the environment and optimizing behavior entirely within that model. However, DreamerV3 treats the robot as an external actor whose body is implicitly encoded in the world model observations, rather than as an entity with an explicit, queryable representation of its own morphology. This limits the agent’s ability to reason about its own physical state, particularly under partial observations or after morphological changes such as damage.

The Free-Form Kinematic Self-Model [3] offers a complementary capability: a lightweight, self-supervised neural model that learns the three-dimensional morphology and kinematics of a robot purely from visual observation and joint angle data, without requiring CAD models or hand-crafted kinematic equations. Its differentiable nature and low memory footprint make it an attractive candidate for integration into a model-based RL pipeline.

This work explores whether combining these two frameworks yields a more capable and robust agent. The implementation builds on NaturalDreamer [4], a clean and accessible reimplementation of DreamerV3 that served as the codebase foundation for this project.

The remainder of this paper is structured as follows. Section 2 introduces the DreamerV3 and FFKSM architectures. Section 3 describes the proposed integration in detail. Section 4 outlines the experimental setup. Section 5 presents and discusses the results. Section 6 concludes with planned directions for future work.

2 Background

2.1 DreamerV3

DreamerV3 [2] is a general-purpose model-based reinforcement learning algorithm that learns a compact latent model of the environment and optimizes agent behavior entirely within that model, without direct environment interaction during policy learning. It outperforms specialized algorithms across more than 150 diverse tasks using a single fixed set of hyperparameters, making it one of the most general RL algorithms to date.

World Model. The core of DreamerV3 is a *Recurrent State-Space Model* (RSSM), which combines a deterministic recurrent state $h_t \in \mathbb{R}^{512}$ with a stochastic latent representation z_t . Together they form the full model state $s_t = \{h_t, z_t\}$. At each step, the sequence model updates h_t based on the previous state and action, while the encoder maps the current observation x_t to the posterior $z_t \sim q_\phi(z_t | h_t, x_t)$. A dynamics predictor learns the prior $\hat{z}_t \sim p_\phi(\hat{z}_t | h_t)$, enabling the model to imagine future states without real observations. From s_t , separate MLP heads predict rewards, episode continuation, and reconstruct the original observations. Encoders and decoders use CNNs for image inputs and MLPs for vector inputs.

The world model is trained end-to-end by minimizing three loss terms: a *prediction loss* for reconstructing observations, rewards, and continuation flags; a *dynamics loss* that trains the prior to match the posterior; and a *representation loss* that makes the posterior more predictable by the prior. Both KL-based losses are clipped below 1 nat (free bits) to prevent degenerate solutions. The stochastic representations z_t are sampled from categorical distributions using straight-through gradients to allow backpropagation through the discrete sampling step.

Actor and Critic. The actor π_θ and critic v_ψ operate exclusively on imagined trajectories of model states generated by the world model, rather than on real environment data. Starting from a real model state, the world model rolls out a sequence of purely imagined future states by repeatedly applying the dynamics predictor, allowing the agent to evaluate the long-term consequences of actions without additional environment interaction.

The critic estimates the distribution of discounted λ -returns — a weighted mixture of multi-step returns and bootstrapped value estimates — using a categorical representation over exponentially spaced bins. This formulation decouples gradient magnitudes from the scale of return targets, enabling stable learning across environments with vastly different reward magnitudes. To provide stable regression targets, the critic is regularized towards an exponential moving average of its own past parameters, avoiding the need for a separately frozen target network.

The actor maximizes expected returns while maintaining exploration through entropy regularization. To allow a single fixed entropy coefficient $\eta = 3 \times 10^{-4}$ across all domains, returns are normalized robustly via an exponential moving average of their empirical 5th-to-95th percentile range. The REINFORCE estimator is used for both discrete and continuous action spaces, with the advantage weighted by the normalized return scale.

Robustness across domains. A central design goal of DreamerV3 is stability across domains with widely varying input and reward scales. The *symlog* transformation $\text{symlog}(x) = \text{sign}(x) \ln(|x| + 1)$ compresses large positive and negative values symmetrically around the origin and is applied to vector observations and decoder targets. For scalar targets such as rewards and returns, the *symexp twohot* loss encodes values as soft distributions over fixed exponentially spaced bins, preventing gradient magnitudes from depending on target scale. Together, these techniques allow the algorithm to operate with a single configuration across tasks as diverse as Atari games, continuous control, and 3D navigation.

2.2 Free-Form Kinematic Self-Model

The Free-Form Kinematic Self-Model (FFKSM) [3] is a self-supervised framework that enables a robot to learn a neural model of its own morphology and kinematics purely from raw video and joint angle data, without CAD models, depth sensors, or manually engineered kinematic equations. The key motivation is to give robots an ability analogous to observing their own body in a mirror: by watching their own movements, they internalize the relationship between joint configurations and three-dimensional body shape.

Architecture. The FFKSM consists of three small neural networks: a *Coordinates Encoder*, a *Kinematic Encoder*, and a *Predictive Module*. Given a 3D query point $X \in \mathbb{R}^3$ and the current joint angles $A \in \mathbb{R}^4$, the model predicts whether that location is occupied by the robot body and whether it is visible to the camera.

To reduce the complexity of the learning problem, the first two joint angles — responsible for yaw and pitch of the base — are factored out via an explicit geometric transformation. The query point X is projected into a virtual coordinate frame $X' = (R_{\text{pitch}} \cdot R_{\text{yaw}})^{-1}X$ in which the base joints are effectively fixed and only the camera appears to move. This separation means the Coordinates Encoder only needs to learn a static spatial embedding, while the Kinematic Encoder processes the remaining distal joints (A_2, A_3) independently. The Predictive Module combines both encoded representations and outputs density and visibility estimates at each queried point. Density values are passed through a ReLU-based activation to ensure positivity and training stability.

Rendering and training. Predicted density and visibility values along each camera ray are accumulated via volumetric rendering — analogous to Neural Radiance Fields (NeRF) — to produce a predicted binary image of the robot. The model is trained through *motor babbling*: the robot executes random joint configurations while a single RGB camera records the resulting motion. Each frame is paired with the corresponding joint angles, and the model is trained by minimizing the MSE between the rendered prediction and a segmented binary ground-truth image. No supervision beyond camera frames and joint angles is required. The resulting model is task-agnostic, requires only 333 KB of memory, and typically converges within a few thousand training steps.

Downstream capabilities. Because the FFKSM is fully differentiable with respect to both spatial query coordinates and joint angles, it enables gradient-based inverse kinematics by optimizing joint angles to place the predicted end-effector at a target position. For motion planning, two FFKSM instances are combined with an RRT planner: one evaluates whole-body collision states, and the other provides end-effector heuristics to guide the search. Finally, physical damage can be detected by measuring the divergence between model predictions and live camera observations. Upon detection, the model can be fine-tuned on post-damage data to recover an accurate morphological representation, enabling autonomous adaptation without human intervention.

3 Methodology

This section describes the architectural integration of the Free-Form Kinematic Self-Model into the DreamerV3 framework. The central idea is to augment the agent’s decision-making with continuous, vision-based morphological self-knowledge by providing the actor, critic, reward predictor, and continue predictor with an additional latent representation derived from the FFKSM.

Parameter	Value
Recurrent state size	512
WM latent size	16x16
SM latent size	32
Full state size	800
SM latent scaling	x10
Warmup steps	500
SM/WM update frequency	every 4 steps
Damage threshold factor	x100

Table 1: Important Hyperparameters

3.1 Integration Overview

In the original DreamerV3 architecture, the actor and all associated prediction heads operate on a *full state* $s_t = \{h_t, z_t\}$, which concatenates the recurrent state $h_t \in \mathbb{R}^{512}$ produced by the sequence model with the stochastic world model latent $z_t \in \mathbb{R}^{256}$ (a 16×16 categorical representation), yielding a full state dimensionality of 768.

In the proposed architecture, the full state is extended with an additional latent vector $\ell_t^{\text{SM}} \in \mathbb{R}^{32}$ extracted from the FFKSM, producing an augmented full state

$$\tilde{s}_t = \{h_t, z_t, \ell_t^{\text{SM}}\} \in \mathbb{R}^{800}. \quad (1)$$

This augmented state $\tilde{s}_t$ is used as input to the actor, critic, reward predictor, and continue predictor in place of the original s_t .

3.2 Extraction of Self-Model Latents

The FFKSM architecture terminates in a predictive module whose intermediate activations are decoded into a rendered output image (e.g., an RGB or binary occupancy map) for loss computation. In the proposed integration, the latent activations that serve as input to the decoder are **directly read out** from the feed-forward predictive module, immediately before the final decoding step, without any additional learned transformation. This extracts a 32-dimensional latent vector ℓ_t^{SM} from the internal representation of the network. The original decoding pathway is preserved so that the FFKSM training loss remains unaffected.

The WM latents z_t are discrete categorical variables sampled via straight-through estimation, and their effective values are consequently close to binary. The SM latents ℓ_t^{SM} , by contrast, are continuous activations with a substantially smaller magnitude. To bring both representations into a comparable numerical range and prevent the SM latents from being suppressed as noise by downstream networks, they are scaled by a constant factor of 10 prior to concatenation:

$$\tilde{\ell}_t^{\text{SM}} = 10 \cdot \ell_t^{\text{SM}}. \quad (2)$$

This scaling factor was determined empirically. The implications of this design choice are discussed further in Section 5.

3.3 Augmented Prediction Heads

Actor and Critic. Both the actor π_θ and the critic v_ψ receive the augmented full state $\tilde{s}_t$ as their sole input. The motivation is straightforward: morphological self-knowledge may inform the agent about the physical feasibility and consequences of candidate actions, providing information that is not captured by the world model latents alone.

Reward Predictor. The reward predictor is likewise conditioned on $\tilde{s}_t$. Since the reward predictor plays a central role in both actor and world model training through the imagined return estimates, it is hypothesized that incorporating morphological information may allow the reward signal to better reflect the physical state of the robot. This hypothesis is evaluated experimentally in Section 5.

Continue Predictor. The continue predictor, which estimates the probability that an episode has not yet terminated, also receives $\tilde{s}_t$. The rationale is that morphological information may carry cues about episode continuation: for instance, a robot approaching a joint limit or an obstacle may be more likely to reach a terminal state, and such configurations are directly reflected in the FFKSM latents.

3.4 Image Preprocessing for the Self-Model

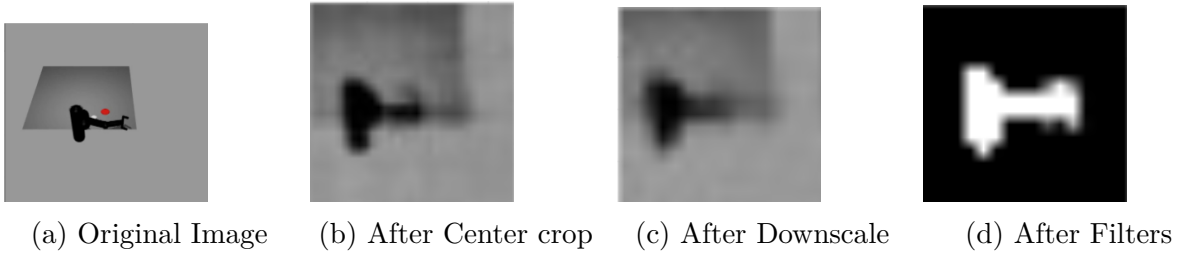


Figure 1: The process of an observation before being used in the FFKSM

The FFKSM operates on binary segmented images of the robot against a uniform background. To prepare the raw camera frames collected during environment interaction, a dedicated preprocessing pipeline is applied:

1. **Center crop:** the image is cropped to a square region centered on the robot, removing irrelevant border content.
2. **Downscaling:** the cropped image is resized to 16×16 pixels to reduce computational load.
3. **Color filtering:** a chroma-key filter isolates the robot body from the background.
4. **Color distortion removal:** residual color artifacts introduced by lighting conditions or sensor noise are suppressed.
5. **Binarization:** the resulting image is converted to a binary occupancy map, where pixels belonging to the robot are set to 1 and background pixels to 0.

This pipeline significantly reduces the memory and runtime requirements of the FFKSM compared to processing full-resolution images. In addition to the preprocessed images, the joint angles A_t are collected as a dedicated data type during every environment interaction step and stored in the replay buffer alongside the standard observation, action, and reward data. The joint angles are read directly from the robot’s motor encoders; in all experiments conducted so far, a simulated robot is used, which makes angle readout straightforward. It is expected that this approach transfers readily to most physical robotic platforms.

Several FFKSM-specific parameters — including camera distance, degrees of freedom, and field of view — must be configured manually before training. These parameters are not required by the standard DreamerV3 pipeline but are also necessary in the original FFKSM framework. Their values are environment-specific and assumed to be readily available for any robotic platform.

3.5 Training Procedure

Warmup phase. At the start of training, a warmup phase of 500 environment steps is executed during which the FFKSM trains but the actor does not. This prevents the actor from learning on SM latents that are still highly uninformative. After approximately 500 steps, the FFKSM has converged sufficiently to capture the coarse morphology of the robot, at which point actor training begins.

Training order within each iteration. Each training iteration proceeds in the following order:

1. **SM training:** the FFKSM is updated using a batch of images and joint angles sampled from the replay buffer.
2. **WM training:** the world model — including the reward predictor and continue predictor — is updated using the same batch. Since both prediction heads now consume the augmented state $\tilde{s}_t$, the SM must be updated first to ensure that the latents $\tilde{\ell}_t^{\text{SM}}$ are current.
3. **Actor and Critic training:** the actor and critic are updated last, as they depend on the complete augmented full state.

Update frequency. To manage the computational cost of the FFKSM, both the SM and the WM are updated only once every four environment steps during imagined rollouts. Crucially, these two updates are synchronized: the SM and WM always train on the same batch in the same step, ensuring that the WM latents z_t and the SM latents $\tilde{\ell}_t^{\text{SM}}$ remain contextually consistent. The actor, being computationally lighter, is updated at every step; it therefore trains on the same latent values for three consecutive steps between each SM/WM update.

During real environment interaction, the SM latent is refreshed at every step.

The FFKSM typically converges within approximately 4000 training steps across all experiments conducted so far, at which point its loss stabilizes and SM training is halted.

3.6 Damage Detection and Recovery

After the FFKSM has converged, a lightweight damage detection mechanism operates continuously during environment interaction. At each step, the current FFKSM loss $\mathcal{L}_{\text{SM}}^{\text{live}}$

is compared against the reference loss $\mathcal{L}_{\text{SM}}^{\text{ref}}$ recorded at the end of training. A damage event is triggered when

$$\mathcal{L}_{\text{SM}}^{\text{live}} > 100 \cdot \mathcal{L}_{\text{SM}}^{\text{ref}}. \quad (3)$$

The threshold factor of 100 was chosen conservatively to avoid false positives: during normal operation, the live loss is consistently higher than the training loss, so a large margin is necessary. This value is provisional and is expected to be refined once dedicated damage detection experiments are conducted.

Upon detection of a damage event, the FFKSM re-enters training mode. Only data collected *after* the damage event is used for fine-tuning, ensuring that the model adapts to the new morphology rather than averaging over pre- and post-damage observations. The SM latents are neither zeroed nor frozen during recovery: since damage typically induces only local morphological changes rather than a complete restructuring, the existing latent representation serves as a useful initialization and accelerates convergence. Actor and critic training likewise continue uninterrupted during recovery, as the agent must keep collecting data to supply the fine-tuning process, and the relatively small morphological change is not expected to destabilize the policy significantly.

4 Experiments

Two experiments were conducted using the **Pusher-v5** environment from the MuJoCo suite [1], complemented by a benchmark run of the NaturalDreamer baseline [4] on the same hardware for direct comparison. In all runs, total reward, world model loss, reconstruction loss, reward predictor loss, KL loss, actor loss, critic loss, entropies, log probabilities, advantages, critic values, and SM loss were recorded.

4.1 Environment

Pusher-v5 features a seven-DOF robotic arm that must push a cylindrical object to a goal position using its end effector. Actions are joint torques in $[-2, 2]$ N m. The reward penalizes fingertip-to-object distance, object-to-goal distance, and control magnitude, making it effectively a penalty signal to be minimized. Both agents receive raw pixel observations; no proprioceptive state is provided to the world model.

4.2 Experiment 1: Shared Camera

Both the baseline and the proposed architecture were trained for 60 000 steps under identical conditions with a single fixed third-person camera. The world model received unmodified RGB frames; the FFKSM received the same frames passed through the preprocessing pipeline (Section 3.4). Joint angles were additionally stored in the replay buffer for FFKSM training. The goal was to assess whether SM latents provide any measurable benefit over the unmodified baseline under equal observational conditions.

4.3 Experiment 2: Decoupled Cameras

The second experiment simulated a more realistic scenario in which the world model receives only an ego-centric view while a separate external camera serves exclusively as the

morphological sensor for the FFKSM. Two cameras were used: an *arm camera* mounted at the top of the robot’s vertical base column, co-rotating with the arm and providing a partial, ego-centric view to the world model; and the original fixed *third-person camera*, accessible only to the FFKSM. This setup is motivated by practical deployments where a retractable auxiliary camera could capture morphology during self-model training while remaining stowed during task execution. The experiment investigated whether SM-encoded morphological knowledge can compensate for the reduced observational context of the world model. Training was stopped after 45 000 steps, as the qualitative outcome was already apparent.

5 Results and Discussion

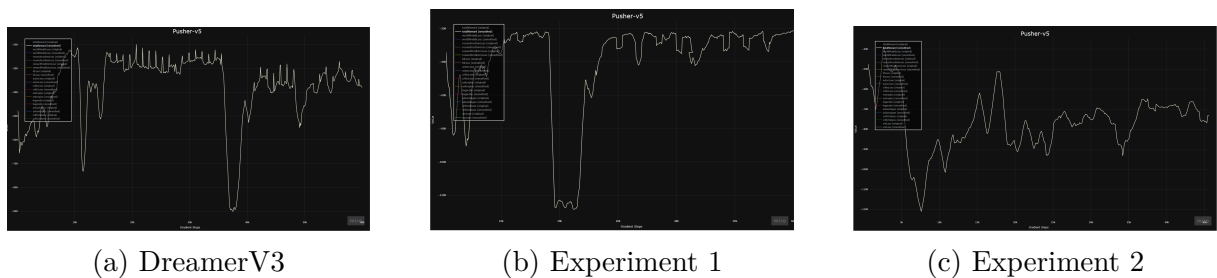


Figure 2: Total Reward

5.1 Benchmark

The NaturalDreamer baseline [4] ran stably for 60 000 steps on the target hardware, confirming that the codebase and environment are correctly configured and providing a hardware-matched reference for the subsequent experiments.

5.2 Experiment 1: Shared Camera

Total reward. Both agents achieved broadly similar overall performance. The proposed architecture reached a mean reward of approximately -250 after recovery, compared to approximately -350 for the NaturalDreamer baseline, suggesting a modest improvement in final policy quality. Both runs exhibited a pronounced reward collapse: the baseline dropped to approximately -900 between steps 35 000 and 40 000 before partially recovering, while the proposed architecture dropped more severely to approximately -1300 between steps 18 000 and 27 000 but recovered fully. The earlier onset and longer duration of the collapse in the proposed architecture may be attributable to the additional complexity introduced by the augmented state $\tilde{s}_t$: the actor must integrate SM latents alongside WM latents, potentially requiring more steps to re-establish a coherent policy after a destabilization event. The collapse itself is not yet fully understood and may reflect a known instability in the DreamerV3 training dynamics.

KL loss. The KL loss followed a broadly similar descending trend in both agents, but the proposed architecture exhibited frequent large spikes that persisted until approximately step 40 000, after which the loss stabilized. In the baseline, spikes were markedly less

pronounced. One plausible explanation is that the SM latents influence the actor’s behavior in ways that are not captured by the world model’s prior-posterior dynamics alone: since the actor now acts on morphological information that is external to the RSSM, the encoder’s posterior can diverge more strongly from the dynamics prior. If correct, this would constitute positive evidence that the SM latents have a genuine effect on agent behavior, rather than being ignored as noise.

Actor loss. The actor loss mirrored the reward trajectory, with a collapse and recovery pattern consistent with the total reward curve. The recovery in the proposed architecture was notably cleaner, with fewer spikes after stabilization, suggesting that the augmented state ultimately supports more stable policy optimization despite the longer initial instability.

SM loss. The SM loss decreased rapidly and stabilized at a low level by approximately 10 000 steps, consistent with the expected convergence behavior described in Section 3.5. A brief temporary increase around step 5 000 likely reflects a scheduled additional training pass by the FFKSM. Overall, the SM converged as expected and did not exhibit pathological behavior.

Scaling and noise interpretation. A key concern identified during this experiment is that the SM latents, even after the $\times 10$ scaling described in Section 3.2, may still be interpreted as noise by the downstream networks rather than as informative features. The WM latents are straight-through categorical variables with near-binary values, whereas the SM latents are continuous activations of substantially smaller and less structured magnitude. Bridging this representational gap remains the central open problem of the current architecture.

5.3 Experiment 2: Decoupled Cameras

Total reward and actor loss. Performance in the decoupled camera setup was substantially worse than in Experiment 1. The total reward was highly unstable throughout training, with frequent large spikes, a best observed value of approximately -400 , and a mean of approximately -700 . The actor loss showed a similar pattern. The ego-centric arm camera provides a rapidly and non-stationarily changing observation stream — particularly because the arm tends to rotate to its mechanical limits in early training, causing large frame-to-frame shifts — which makes it significantly harder for the world model to learn a stable latent representation.

WM loss. The world model loss was initially high and unstable, then abruptly stabilized at a low value around step 17 000. This transition coincided precisely with the moment at which the total reward reached its best observed spike. A plausible interpretation is that the arm briefly adopted a more stationary posture above the platform at that point, providing the ego-centric camera with a more consistent sequence of frames and allowing the world model to consolidate its representations. This remains speculative, as no direct behavioral evidence was recorded.

Overall assessment. The results of both experiments suggest that the current integration does not yet achieve the intended goal. The SM latents do not demonstrably improve task performance, and the representational mismatch between continuous SM activations and discrete WM latents is likely preventing the actor from extracting useful morphological information. The decoupled camera setup further compounds this by introducing a more challenging observation problem that the architecture is not yet equipped to solve.

6 Future Work

Several directions are planned to address the limitations identified above.

The most immediate priority is resolving the representational mismatch between SM and WM latents. A promising approach is to learn a small encoder network that maps the continuous SM activations into a discrete categorical space compatible with the WM latents, making the morphological information structurally consistent with the representations the actor already relies on.

A second direction reconceptualizes the role of the self-model within the architecture: rather than providing SM latents as an additional input to the actor, they would be fed directly into the world model, treating the robot’s own morphology as part of the world to be modeled. In a simulated baseline setting, this is not expected to yield dramatic improvements over standard DreamerV3, but maintaining stable and comparable performance would already be a satisfying outcome. The deeper motivation is real-world applicability: in physical deployments, a persistent global third-person camera is rarely available, and the world model must typically operate from a first-person view alone. In such settings, an integrated morphological self-model could provide the world model with the body-awareness it would otherwise lack, enabling more robust planning under egocentric and partial observations.

Finally, once the base architecture is stabilized, dedicated damage detection experiments will be conducted to empirically calibrate the detection threshold introduced in Equation 3 and to evaluate the recovery mechanism under controlled morphological perturbations.

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